

Nguyen Gia Huy | AI Engineer Intern

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Professional Summary

A 3rd-year IT student pursuing an AI Engineer career path with a solid foundation in Computer Vision, Deep Learning, and Generative AI. Implemented models for Object Detection, Image Processing, and model optimization through scientific research (published at STAIS 2025). Proficient in Python and frameworks such as PyTorch and TensorFlow. Capable of managing the entire data pipeline (Preprocessing, Augmentation, Labeling). Skilled in building REST APIs (FastAPI) and containerizing models with Docker for production environments. Eager to find an internship opportunity to contribute value to real-world AI projects and further develop professional skills in a corporate environment.

Education

University of Transport Ho Chi Minh City

Information Technology – GPA: 3.4/4.0

Expected Graduation: 2026

Projects

QBCA – Automated Question Bank Deduplication System

May 2025 – Aug 2025

Technologies: Python, Pytorch, BERT, FAISS, Cosine Similarity, FastAPI

- **Objective:** Replace manual verification with AI to automatically detect semantically duplicated questions (Semantic Similarity), optimizing educational data management.
- **Roles & Techniques:** Utilized BERT for text embedding, stored vectors in FAISS to optimize retrieval and scalability for large datasets, applied Cosine Similarity to calculate similarity scores and filter duplicates.
- **Results:** Achieved 96% accuracy on 400 test cases, successfully deployed the AI Service using FastAPI for flexible integration, ensuring stable and scalable system operation.

Real-time Driver Fatigue Detection System

Sep 2025 – Dec 2025

Technologies: Python, YOLO, MediaPipe, OpenCV, PyTorch

- **Objective:** Early detection of driver drowsiness and distraction (yawning, nodding off) to provide real-time alerts and ensure traffic safety.
- **Roles & Techniques:** Fine-tuned YOLO on a dataset of 3,000 images for yawn detection, used MediaPipe to extract facial landmarks and head pose, calculating the EAR (Eye Aspect Ratio) index for drowsiness detection. Applied pHash combined with LRU Cache to identify similar frames, reducing redundant YOLO model calls.
- **Results:** Achieved 92% accuracy under good lighting conditions and significantly optimized processing speed thanks to LRU Cache.

Skills

- **AI/Computer Vision:** PyTorch, TensorFlow, YOLO, OpenCV, Image Processing, Object Detection/Classification.
- **Generative AI & NLP:** LLMs, BERT, GANs, Transformers.
- **Data Engineering:** Data Preprocessing, Data Augmentation.
- **Languages & Backend:** Python, SQL (MySQL, SQL Server), FastAPI, RESTful API.
- **Tools & MLOps:** Git, Docker, Linux, W&B, MLflow.

Publications

SPD-GAN: Difficulty Adjustment for Discriminators in Generative Models

May 2024 – Aug 2025

Main Author | National Scientific Conference STAIS 2025

- Proposed a mathematical framework for adjusting Discriminator difficulty to stabilize training and mitigate Mode Collapse.
- Experiments on CIFAR-10 showed a 39% increase in quality and 11% increase in diversity of generated images while reducing Mode Collapse.
- Accepted and published at the National Scientific Conference STAIS 2025.
- **Paper link:** https://drive.google.com/file/d/18fGkvpt8ttuyX-kumGp7_obACivowqAy/view?usp=sharing

Applying Machine Learning to Analyze Delivery Time Factors and Customer Reviews *Apr 2025 – May 2025*

Main Author / National Scientific Conference CTST 2025

- Applied Lasso/Ridge Regression to mitigate multi-collinearity and Overfitting in Logistics data.
- Conducted in-depth Feature Engineering: processed time-series and distance data, utilized Lasso for feature selection.
- Published at the National Scientific Conference CTST 2025.

Languages

- **English:** B2 level.
- **Chinese:** HSK 3.

Awards

- Google Student Ambassador – Selected as Top 700 Representatives Nationwide (2026).
- Bronze Medal, 12th Applied Manufacturing Design Competition (2024).
- 2nd Prize, 8th Food Safety & Security Conference – AI in Nutrition Management (2024).
- 3rd Prize, Blockchain in Food Traceability (2024).